

The bacterium *Clostridium difficile* secretes protein toxins TcdA and TcdB, two homologues associated with *C. difficile*–related colitis. Tcd proteins enter host intestinal cells via endocytosis induced by interactions between the Tcd C-terminal domain and cell surface receptors. Endocytosis is followed by translocation of the N-terminal glucosyltransferase domain (GTD) across the endosomal membrane. The GTD remains anchored to endosomes by the central region of the protein, a transmembrane domain believed to form membrane pores and facilitate translocation. Inositol hexakisphosphate (IP6) binds Tcd proteins, an interaction that is stabilized by salt bridges in the binding pocket (Figure 1).

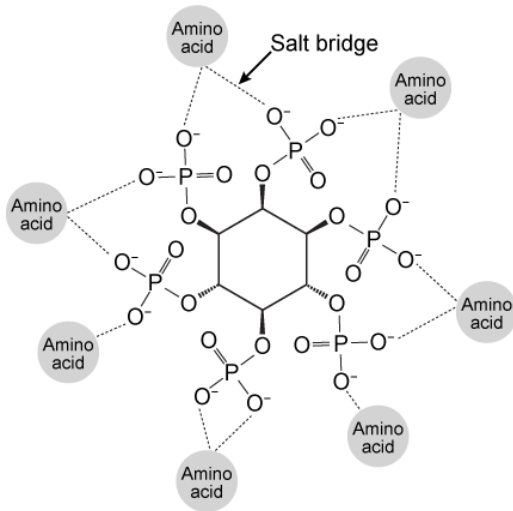


Figure 1 Salt bridges formed between IP6 and the amino acids in the binding pocket

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To better understand the relationship between IP6 binding and catalytic activity, autoprocessing assays were performed. Researchers mutated histidine to alanine at positions 759 and 757 in the flap regions of TcdA and TcdB, respectively. They then compared the autoprocessing activities of H759A and H757A mutants to those of wild-type TcdA and TcdB at varying concentrations of IP6 (Figure 2).

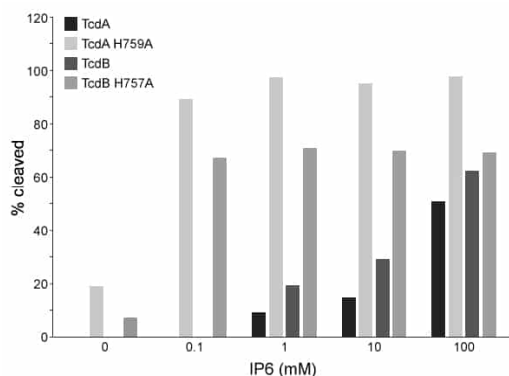


Figure 2 Percent cleavage of wild-type and mutant Tcd by autoprocessing

The first step of the autoprocessing reaction mechanism described in the passage involves modification of a thiol in the active site by an amino acid functioning as a general base. This action

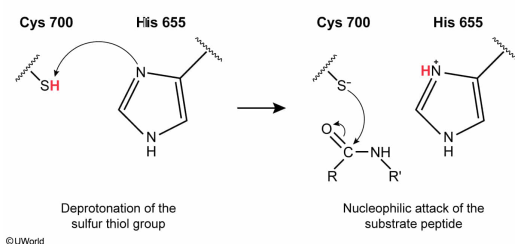
is likely to enhance protease activity by:

- ✔ ☒ A. the deprotonation of cysteine, which results in an anionic sulfur with increased nucleophilicity. [70%]
- ☐ B. the protonation of serine, which results in a structure with improved capacity to act as a leaving group. [6%]
- ☐ C. the deprotonation of serine, which results in an anionic oxygen with increased nucleophilicity. [8%]
- ☐ D. the protonation of cysteine, which results in a structure with improved capacity to act as a leaving group. [14%]

Correct

70% answered correctly

Explanation:



In general, amino acids containing thiol ($-\text{SH}$) or hydroxyl ($-\text{OH}$) groups **can act as nucleophiles** in biological reactions. Nucleophiles are attracted to the positive charge of atomic nuclei, and the greater the negative charge possessed by a molecule, the greater its nucleophilicity.

Deprotonation of the **cysteine thiol** or the **hydroxyl group** of serine results in negatively charged sulfur or oxygen atoms and **enhances** their otherwise weak **nucleophilicity**.

The question states that the mechanism begins with the modification of a thiol group. **Cysteine** is the only amino acid with a thiol ($-\text{SH}$) in its R group. The question further states that the thiol is modified by an amino acid (histidine, in this case) acting as a general base. **Bases** are **proton acceptors**; therefore, histidine must **deprotonate** (not protonate) cysteine.

The deprotonation of cysteine would likely enhance the rate of hydrolysis by converting a cysteine residue in the active site into an anionic—and more nucleophilic—sulfide residue. This sulfide can donate electrons to the carbonyl carbon of the peptide in a process called nucleophilic attack, ultimately leading to hydrolysis. This class of enzyme, known as a cysteine protease, is commonly employed in protein degradation.

(Choices B and D) A general base will deprotonate (rather than protonate) a reaction partner. Although protonation would produce a good leaving group, it would not aid in attacking the carbonyl of a peptide bond.

(Choice C) Deprotonation of serine would produce a good nucleophile, and serine proteases are common in nature. However, serine does not have a thiol group.

Educational objective:

Cysteine and serine frequently act as nucleophiles in enzyme-catalyzed reactions, but they are relatively weak nucleophiles on their own. Therefore, the mechanisms of these reactions often include a deprotonation step to enhance nucleophilicity.

Time spent:0 sec

Subject:

Biochemistry

QID:400365

System:

Amino Acids and Proteins

The bacterium *Clostridium difficile* secretes protein toxins TcdA and TcdB, two homologues associated with *C. difficile*–related colitis. Tcd proteins enter host intestinal cells via endocytosis induced by interactions between the Tcd C-terminal domain and cell surface receptors. Endocytosis is followed by translocation of the N-terminal glucosyltransferase domain (GTD) across the endosomal membrane. The GTD remains anchored to endosomes by the central region of the protein, a transmembrane domain believed to form membrane pores and facilitate translocation. Inositol hexakisphosphate (IP6) binds Tcd proteins, an interaction that is stabilized by salt bridges in the binding pocket (Figure 1).

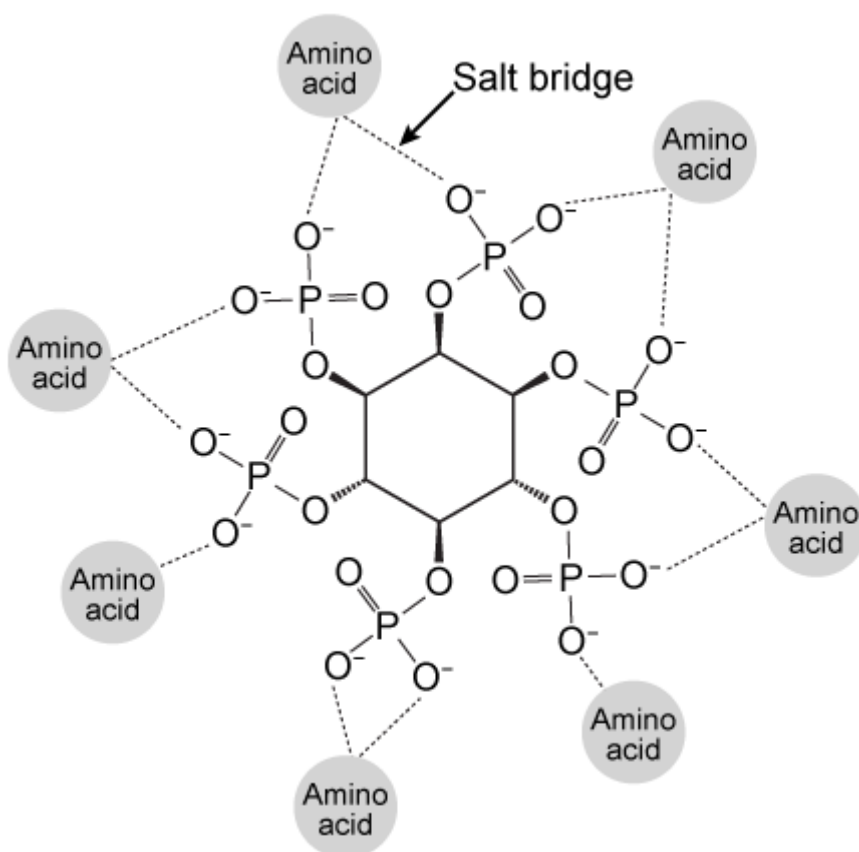


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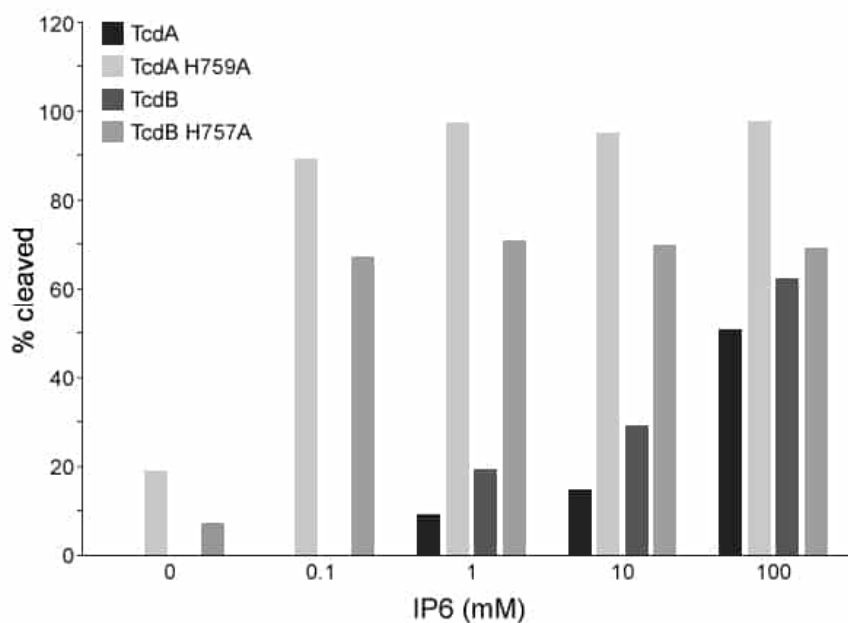


Figure 2 Percent cleavage of wild-type and mutant Tcd by autoprocessing

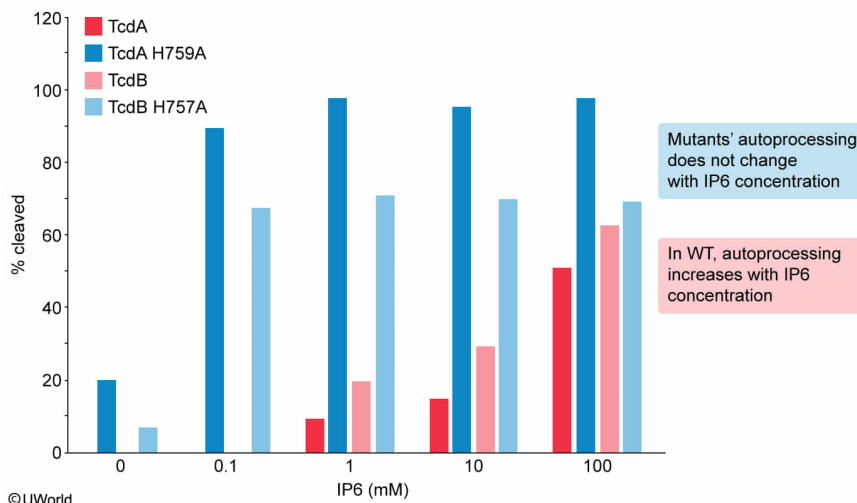
Which conclusion about the effect of the H759A and H757A mutations is best supported by Figure 2?

- ☐ A. Mutant Tcd requires greater concentrations of IP6 to undergo autoprocessing. [6%]
- ✓ ☒ B. Tcd mutant protease activity is less dependent on IP6 than is wild-type activity. [80%]
- ☐ C. Alanine helps control the rate of autoprocessing. [8%]
- ☐ D. Tcd mutants have reduced affinity for IP6. [3%]

Correct

80% answered correctly

Explanation:



In Figure 2, the data show the extent of Tcd autoprocessing that occurs under various conditions. In the absence of IP6, autoprocessing does not occur in the wild-type proteins but does take place in the mutant proteins TcdA H759A and TcdB H757A. When 0.1 mM IP6 is added, the

extent of Tcd hydrolysis increases substantially in mutant proteins but cannot be further increased by adding more IP6, demonstrating that only a small amount of IP6 is required to reach full activity in mutants. In contrast, wild-type autoprocessing steadily increases with increasing IP6 concentration over the range tested. Therefore, the protease activity of TcdA H759A and TcdB H757A is **less dependent on IP6** than the activities of their wild-type counterparts.

(Choice A) Mutant Tcd does not require IP6 to undergo autoprocessing whereas wild-type Tcd does.

(Choice C) The data suggest that histidine at positions 759 and 757 of wild-type TcdA and TcdB, respectively, helps control autoprocessing rates. Substitution of alanine at these positions results in enzymes that are essentially always active, so alanine removes IP6-dependent control of autoprocessing.

(Choice D) If the mutations caused a reduced affinity for IP6, activity would decrease in mutants relative to wild-type at all IP6 concentrations. Instead, mutants have greater activity than wild-type proteins.

Educational objective:

Amino acid substitutions can enhance or diminish enzyme activity. These changes can be tested by measuring protein activity under different conditions.

Time spent:0 sec

Subject:

QID:400366

System:

The bacterium *Clostridium difficile* secretes protein toxins TcdA and TcdB, two homologues associated with *C. difficile*-related colitis. Tcd proteins enter host intestinal cells via endocytosis induced by interactions between the Tcd C-terminal domain and cell surface receptors. Endocytosis is followed by translocation of the N-terminal glucosyltransferase domain (GTD) across the endosomal membrane. The GTD remains anchored to endosomes by the central region of the protein, a transmembrane domain believed to form membrane pores and facilitate translocation. Inositol hexakisphosphate (IP6) binds Tcd proteins, an interaction that is stabilized by salt bridges in the binding pocket (Figure 1).

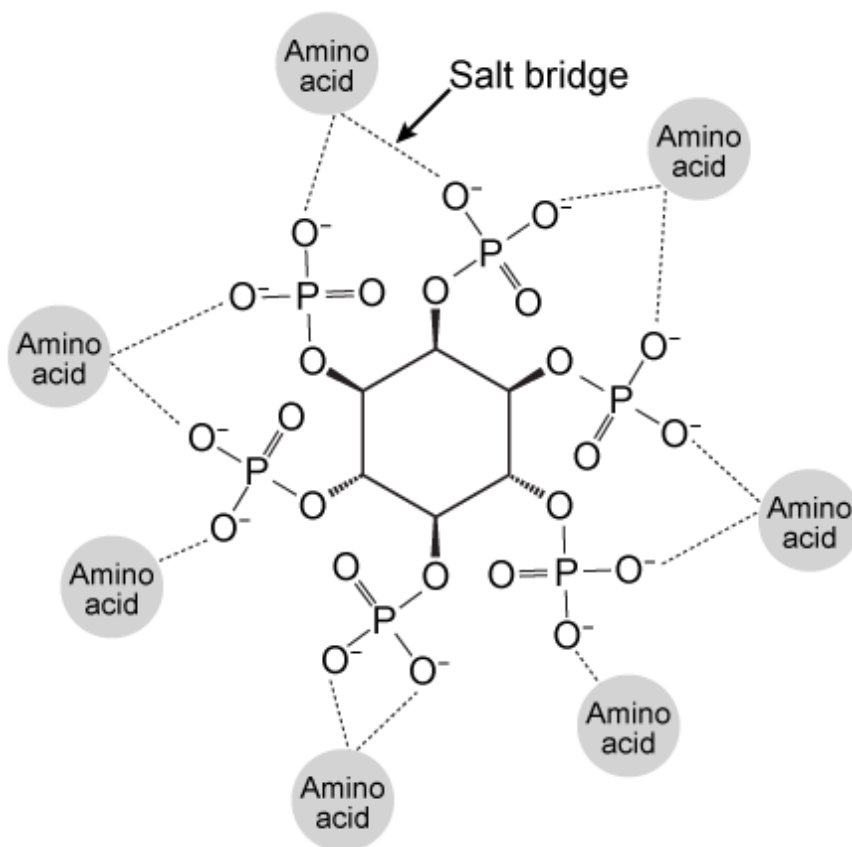


Figure 1 Salt bridges formed between IP6 and the amino acids in the binding pocket

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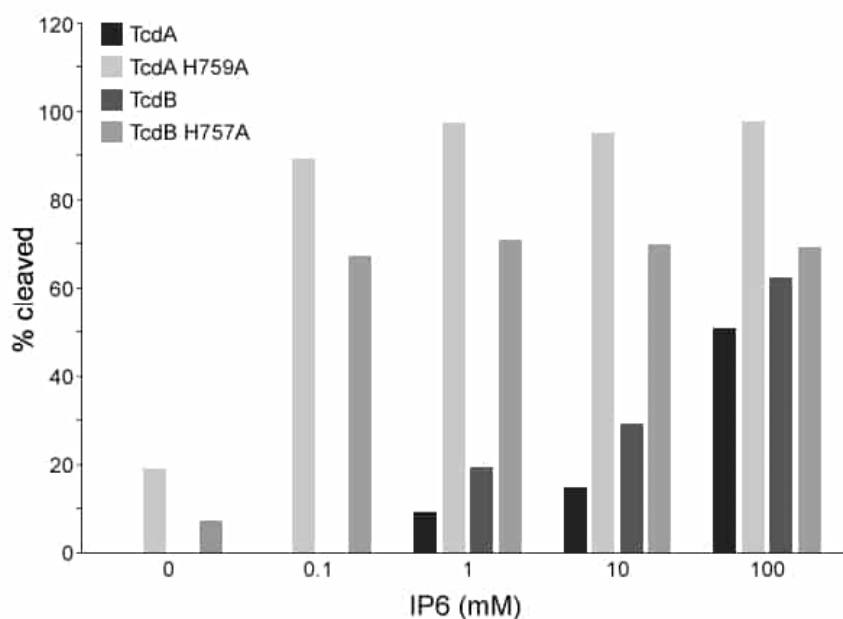


Figure 2 Percent cleavage of wild-type and mutant Tcd by autoprocessing

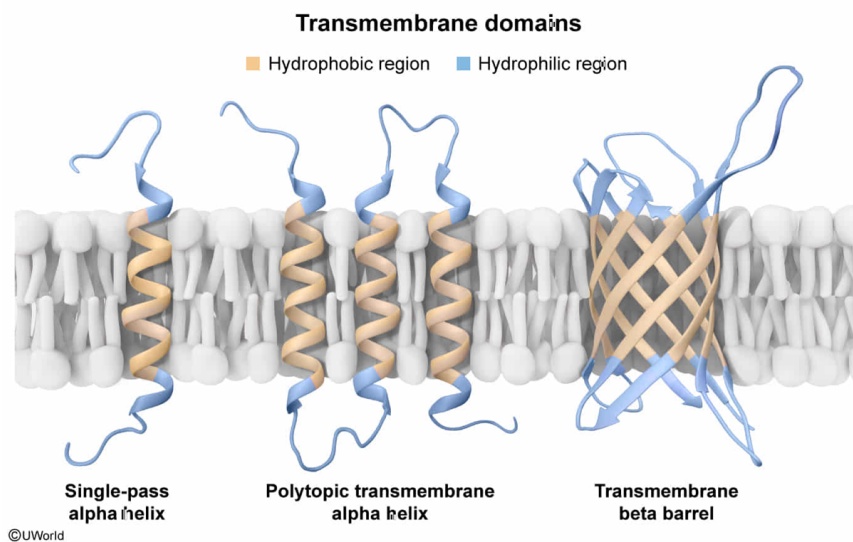
Which of the following amino acid residues is LEAST likely to be present in the Tcd central region?

- ☐ A. Ala [6%]
- ✓ ☒ B. His [72%]
- ☐ C. Val [6%]
- ☐ D. Met [15%]

Correct

71% answered correctly

Explanation:



The central region of TcdA and TcdB is a **transmembrane domain** that anchors the protein to the endosomal membrane. Transmembrane regions span cell membranes one or more times, and often take on α -helical or transmembrane β -barrel arrangements. Membranes consist of phospholipid bilayers composed of inner and outer leaflets of phospholipids oriented with their hydrophilic head groups facing the aqueous environments

and hydrophobic tails facing each other. Because any **hydrophilic amino acid residues** within the membrane would **interact unfavorably** with **hydrophobic phospholipid tails**, only membrane-spanning protein domains containing principally **hydrophobic residues** are thermodynamically stable.

Histidine contains a basic, hydrophilic side chain. As such, histidine residues are most stable in aqueous environments. Of the residues listed, **histidine** would be the **least likely** to be found within a transmembrane domain.

(Choice A) The side chain of **alanine** is a hydrophobic methyl group and interacts well with phospholipid tails.

(Choice C) The side chain of **valine** is a hydrophobic isopropyl group commonly found in transmembrane regions.

(Choice D) The largely aliphatic side chain of **methionine** is relatively hydrophobic, and is frequently found in transmembrane regions.

Educational objective:

Transmembrane proteins contain hydrophobic transmembrane domains composed largely of nonpolar amino acids. They interact favorably with the hydrophobic tails of phospholipids in cellular membranes.

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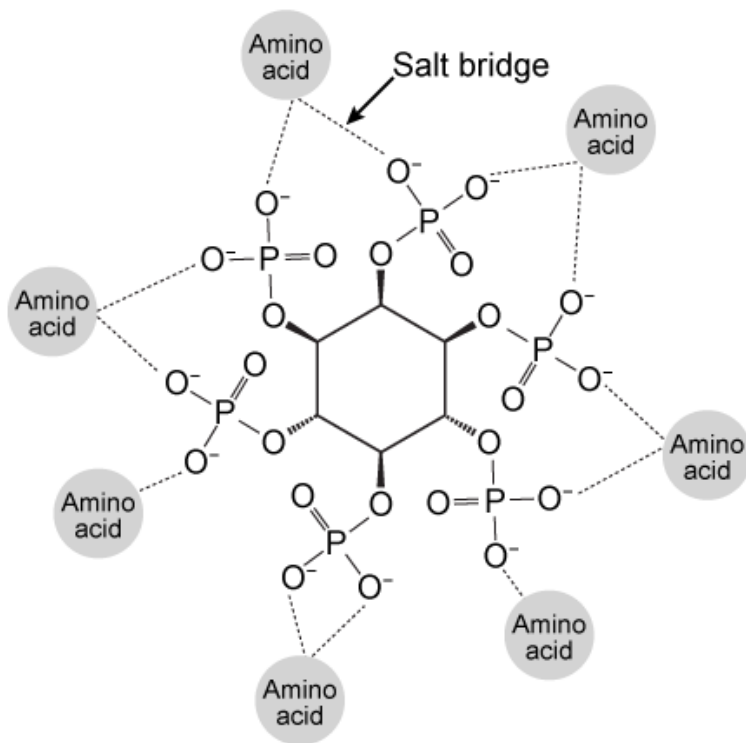


Figure 1 Salt bridges formed between IP6 and the amino acids in the binding pocket

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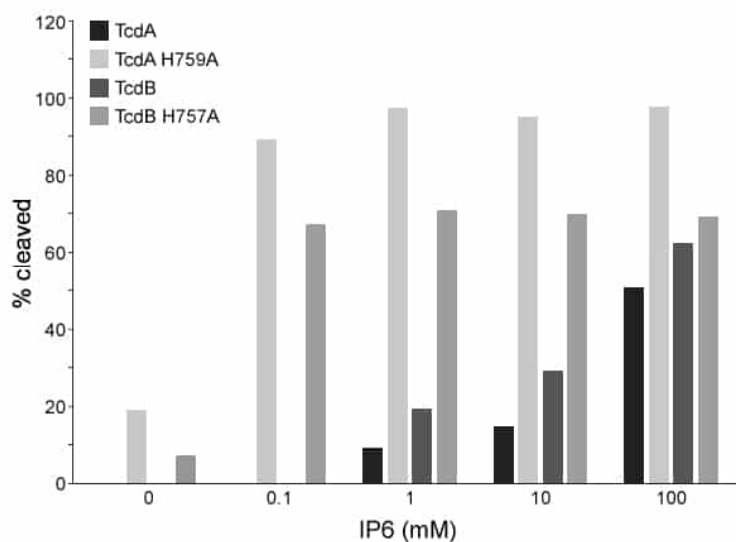


Figure 2 Percent cleavage of wild-type and mutant Tcd by autoprocessing

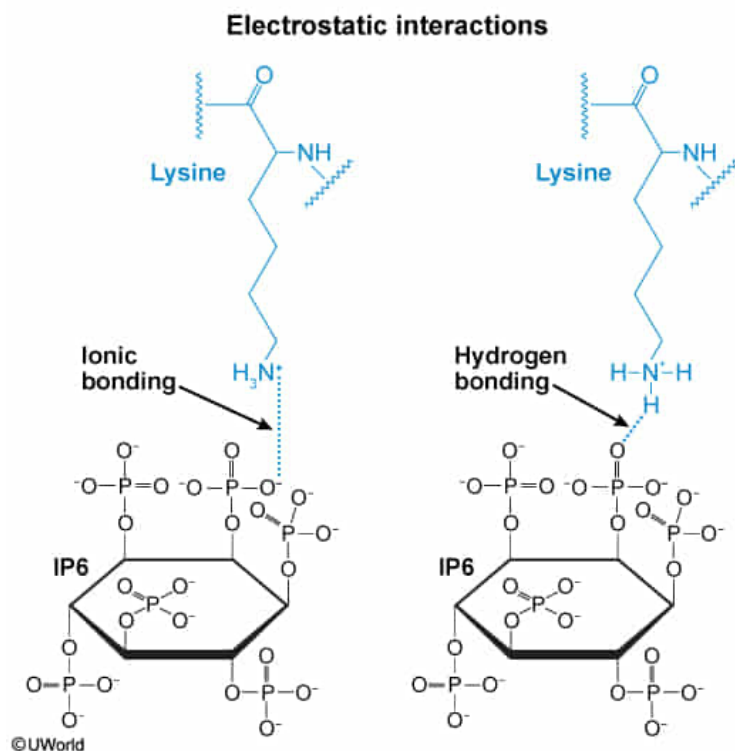
Based on information given in the passage, stabilizing interactions are most likely to occur between phosphate groups on IP6 and which of the following amino acids lining the Tcd binding pocket?

- ✔ ☒ A. Lys⁷⁰⁴ via electrostatic interactions [64%]
- ☐ B. Gly⁷⁰⁵ via hydrophobic interactions [3%]
- ☐ C. Cys⁷⁰⁷ via covalent bonding [15%]
- ☐ D. Asp⁷⁰⁹ via ionic interactions [16%]

Correct

64% answered correctly

Explanation:



The passage states that IP6 is stabilized in the binding pocket through **salt bridging**. In protein chemistry, salt bridging describes an **electrostatic interaction** that occurs between charged residues and ligands, stabilizing the protein conformation or strengthening protein-ligand interactions.

Electrostatic interactions combine elements of both **hydrogen bonding** and **ionic bonding**. In proteins, it most commonly occurs between the anionic **carboxylate of aspartic or glutamic acids** and the cationic **ϵ -amino group of lysine** or the **guanidinium group of arginine** but can potentially involve any pair of oppositely charged chemical groups.

IP6 contains six phosphate groups, each bearing a -2 charge at physiological pH. It can only provide the anionic component of salt bridges; therefore, IP6 must form salt bridges with cationic amino acids. Of the amino acids given, IP6 could potentially form a **salt**

bridge with a cationic lysine residue.

(Choice B) Glycine is neutral and nonpolar, and therefore cannot participate in salt bridge formation. Furthermore, salt bridging is not a hydrophobic interaction.

(Choice C) Although cysteine residues can form covalent disulfide bonds with other cysteine residues, they do not participate in salt bridge formation. Salt bridging is a noncovalent interaction.

(Choice D) Aspartate residues can participate in ionic interactions such as salt bridges but are negatively charged. IP6 can form a salt bridge only with a positively charged residue.

Educational objective:

Salt bridges are electrostatic attractions that contribute to the stability of folded proteins and interactions between proteins and ligands or substrates. Salt bridging combines elements of hydrogen bonding and ionic bonding, and occurs between oppositely charged moieties.

Time spent:0 sec

Subject:
Biochemistry

QID:400368

System:
Amino Acids and Proteins

The bacterium *Clostridium difficile* secretes protein toxins TcdA and TcdB, two homologues associated with *C. difficile*–related colitis. Tcd proteins enter host intestinal cells via endocytosis induced by interactions between the Tcd C-terminal domain and cell surface receptors. Endocytosis is followed by translocation of the N-terminal glucosyltransferase domain (GTD) across the endosomal membrane. The GTD remains anchored to endosomes by the central region of the protein, a transmembrane domain believed to form membrane pores and facilitate translocation. Inositol hexakisphosphate (IP6) binds Tcd proteins, an interaction that is stabilized by salt bridges in the binding pocket (Figure 1).

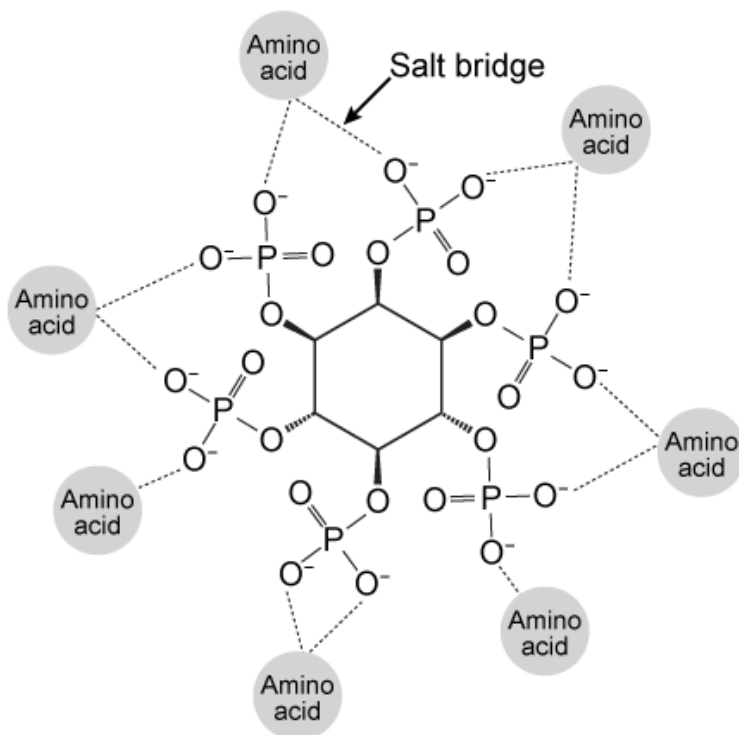


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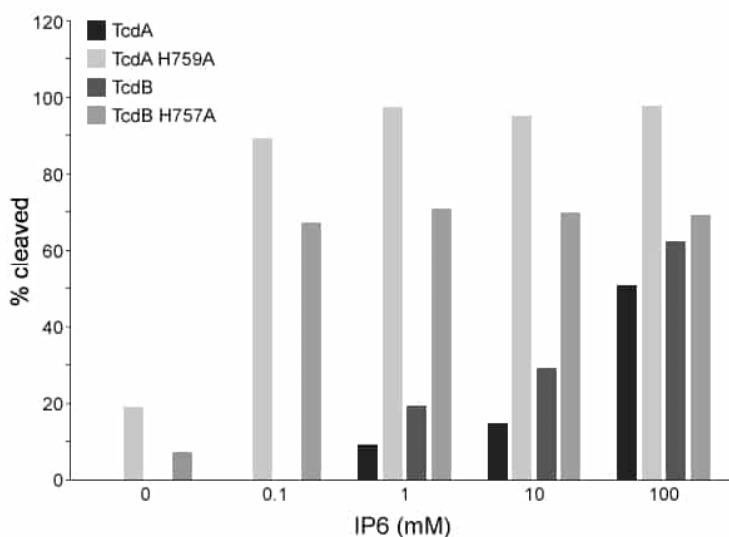


Figure 2 Percent cleavage of wild-type and mutant Tcd by autoprocessing

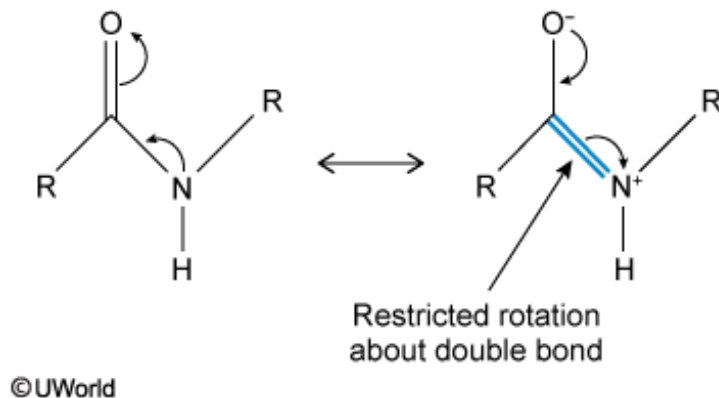
Which of the following statements correctly characterizes the bond that is broken during the reaction catalyzed by the IP6-dependent Tcd protease?

- ☐ A. The bond rapidly degrades in the absence of an enzyme. [10%]
- ✓ ☒ B. The bond restricts molecular rotation. [56%]
- ☐ C. The bond is part of a tetrahedral structure. [23%]
- ☐ D. The nitrogen involved in the bond is negatively charged. [8%]

Correct

55% answered correctly

Explanation:



Proteases are a type of enzyme that catalyze the **hydrolysis of peptide bonds** in polypeptide chains. Peptide bonds are synthesized from a **condensation reaction** between the α -carboxyl group of one amino acid and the α -amino group of another. The amide nitrogen has a lone electron pair that it can donate to the peptide bond, allowing peptide bonds to exist in two **resonance structures** and giving them significant **double-bond character**. Just as a wheel cannot rotate about two axles, atoms in a molecule cannot rotate about two bonds. Therefore, **peptide bonds have restricted rotation**.

The passage states that Tcd undergoes autoprocessing in the form of intramolecular cleavage; in other words, the Tcd protein cleaves itself. In addition, the question asks about a reaction catalyzed by **protease activity**. Therefore, the reaction must be the cleavage of a peptide bond, which has restricted rotation.

(Choice A) Although peptide bond hydrolysis is thermodynamically favorable, it is kinetically slow in the absence of a protease, so peptide bonds do not degrade rapidly in the absence of an enzyme.

(Choice C) The double-bond character of peptide bonds restricts

each peptide unit to a single plane. Therefore, peptide units are planar rather than tetrahedral.

(Choice D) The nitrogen atom in a peptide bond carries a partial *positive* charge due to resonance.

Educational objective:

The peptide bond is a resonance-stabilized, planar structure exhibiting partial double-bond character. Rotation about the bond axis is restricted.

Time spent:0 sec

Subject:
Biochemistry

QID:400369

System:
Amino Acids and Proteins

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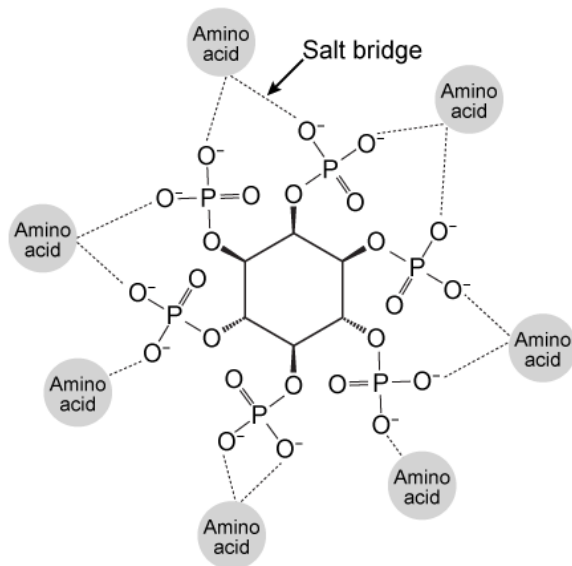


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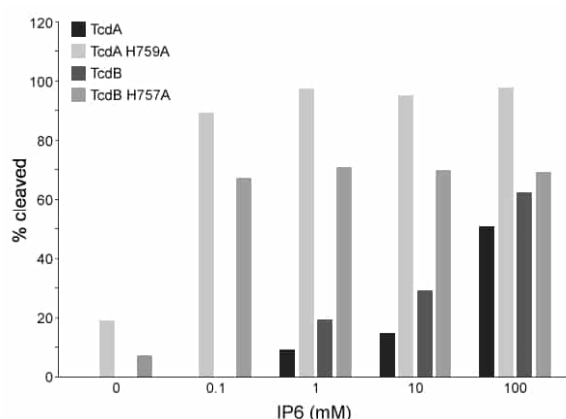


Figure 2 Percent cleavage of wild-type and mutant Tcd by autoprocessing

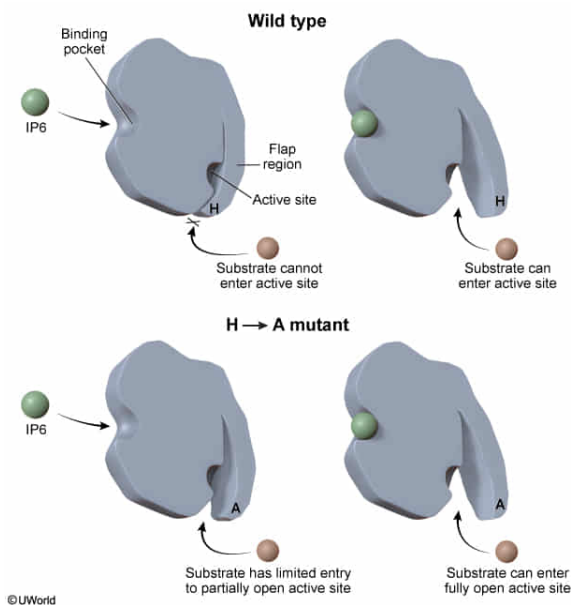
Based on Figure 2, residues H759 and H757 of wild-type TcdA and TcdB, respectively, may function to:

- ☐ A. interact with IP6, stabilizing its interaction in the binding pocket. [28%]
- ☐ B. act as a general base in the IP6-dependent Tcd protease reaction mechanism. [11%]
- ✓ ☒ C. mediate an interaction in the flap region that obstructs the active site in the absence of IP6. [44%]
- ☐ D. create charge repulsion, opposing interaction between the flap region and the active site. [15%]

Correct

44% answered correctly

Explanation:



The passage states that in wild-type Tcd proteins, IP6 binding induces a **conformational change** that moves the flap region away from the active site. This change is an example of **allostery**, a phenomenon in which the binding of an **allosteric effector** (eg, IP6) at an allosteric site (eg, the binding pocket) alters the activity at the catalytic site. Allosteric regulation occurs due to conformational changes induced in the protein upon binding of the effector and can include changes in the conformation of the active site itself, obstruction of the active site or, as in this case, exposure of the active site when the flap region moves.

Unlike wild-type Tcd, autoprocessing occurs in H759A and H757A mutants even in the absence of IP6. This behavior supports the conclusion that in wild-type proteins, an interaction between the flap region and the active site **is abolished** when histidine is mutated to alanine in the mutants. Therefore, H759 and H757 of wild-type Tcd proteins

likely help mediate an interaction that is necessary for the flap region to obstruct the active site in the absence of IP6.

(Choice A) The H → A mutations studied here result in some autoproducting activity even when IP6 is absent. If these histidine residues were required for interaction with IP6, substituting them would decrease activity even at high concentrations of IP6.

(Choice B) If the substituted histidines were acting as bases in the catalytic mechanism, mutating them to alanine would downregulate autoproducting. Instead, the mutation results in increased activity.

(Choice D) The H757A and H759A mutations involve replacing polar, charged histidine with neutral, nonpolar alanine. If histidine were opposing flap-active site interaction via charge repulsion, replacing it with an uncharged amino acid would reduce repulsion and facilitate interaction, decreasing enzymatic activity.

Educational objective:

Allosteric regulation occurs when an allosteric effector binds to the allosteric site and alters catalytic activity in the active site. Mutations that result in the instability of normal conformations can disrupt allosteric regulation.

Time spent:0 sec

Subject:
Biochemistry

QID:400370

System:
Amino Acids and Proteins